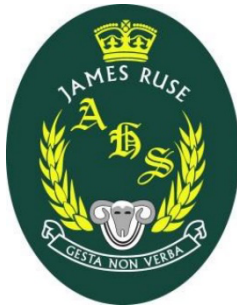


Student Name: _____

Maths class: _____



James Ruse Agricultural High School

2021 YEAR 12 Task 3 Examination

Mathematics Extension 1

General Instructions

Total marks: 40

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 6–12, show relevant mathematical reasoning and/ or calculations

Section I – 5 marks (pages 2–3)

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 35 marks (pages 4–7)

- Attempt Questions 6–12
- Allow about 53 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1. P, Q and R are three collinear points with position vectors $\underline{p}$, $\underline{q}$ and $\underline{r}$ respectively, where Q lies between P and Q. If $|\overrightarrow{QR}| = \frac{1}{2} |\overrightarrow{PQ}|$, then $\underline{r}$ is equal to:

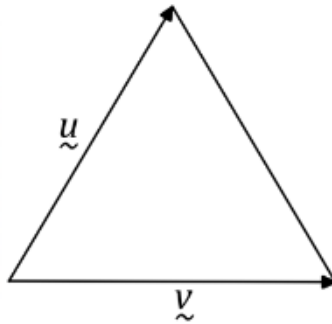
(A) $\frac{3}{2}\underline{q} - \frac{1}{2}\underline{p}$

(B) $\frac{3}{2}\underline{p} + \frac{1}{2}\underline{q}$

(C) $\frac{3}{2}\underline{q} - \frac{3}{2}\underline{p}$

(D) $\frac{1}{2}\underline{p} - \frac{3}{2}\underline{q}$

2. An equilateral triangle of side length 4 units is shown below.



Which of the following is the value of $\underline{u} \cdot \underline{v}$?

(A) 4

(B) $4\sqrt{3}$

(C) $8\sqrt{3}$

(D) 8

3. Which of the following is a well-defined expression which can be calculated involving dot-products for 2-dimensional vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$?

- (A) $(\underline{a} \cdot \underline{b}) \cdot (\underline{b} \cdot \underline{a})$
- (B) $(\underline{a} \cdot \underline{b})\underline{c}$
- (C) $\underline{a} \cdot (\underline{b}\underline{c})$
- (D) $(\underline{a} \cdot \underline{b}) \cdot \underline{c}$

4. The coefficient of $\frac{1}{x}$ in the expansion of $(1+x)^n(1+\frac{1}{x})^n$ is

- (A) $\frac{n!}{(n-1)!(n+1)!}$
- (B) $\frac{(2n)!}{(n-1)!(n+1)!}$
- (C) $\frac{(2n)!}{(2n-1)!(2n+1)!}$
- (D) $\frac{(n+1)!}{2n!(n-1)!}$

5. Let $\underline{p}$, $\underline{q}$, $\underline{r}$ and $\underline{s}$ be four distinct vectors with $\underline{r} \parallel \underline{p}$ and $\underline{s} \parallel \underline{q}$. Which of the following is not an equivalent expression for the vector projection of $\underline{p}$ onto $\underline{q}$?

- | | |
|--|--|
| (A) $\left(\frac{\underline{p} \cdot \underline{q}}{ \underline{q} ^2}\right) \underline{q}$ | (C) $\left(\frac{\underline{p} \cdot \underline{s}}{ \underline{s} ^2}\right) \underline{s}$ |
| (B) $\left(\frac{\underline{p} \cdot \underline{q}}{ \underline{q} }\right) \hat{\underline{q}}$ | (D) $\left(\frac{\underline{r} \cdot \underline{s}}{ \underline{s} ^2}\right) \underline{s}$ |

Section II begins on the next page

Section II

50 marks

Attempt Questions 6-12

Allow about 53 minutes for this section

Start each question on a new page. Extra paper is available.

In Questions 6-10, your responses should include relevant mathematical reasoning and/ or calculations.

Question 6 (5 marks) Start a new page.

- (a) The point A has position vector $5\mathbf{i} - 4\mathbf{j}$ and the point B has position vector $-5\mathbf{i} - 2\mathbf{j}$.
- (i) Find the vector $\overrightarrow{AB}$ 1
- (ii) Find $|\overrightarrow{AB}|$. Give your answer as a simplified surd. 1
- (b) Let P be the point $(5, -8)$ in the cartesian plane.
- (i) Draw the position vector $\overrightarrow{OP}$ 1
- (ii) Write $\overrightarrow{OP}$ as a column vector. 1
- (iii) Find a unit vector in the direction of $\overrightarrow{OP}$. 1

Question 7 (5 marks) Start a new page.

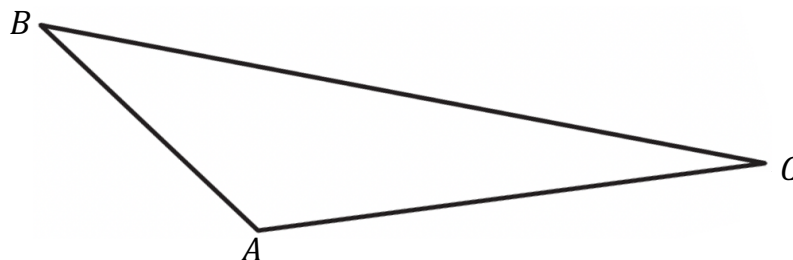
- (a) Find the angle between the vectors $3\mathbf{i} + \mathbf{j}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. 2
- (b) Show that the vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ are perpendicular. 1
- (c) Find a vector of magnitude 15 in the direction of the vector $5\mathbf{i} + 12\mathbf{j}$. 2

Please turn over

Question 8 (5 marks) Start a new page.

- (a) In the triangle ABC below, $\overrightarrow{AB} = -2\mathbf{i} + 7\mathbf{j}$ and $\overrightarrow{AC} = 12\mathbf{i} - \mathbf{j}$.

2



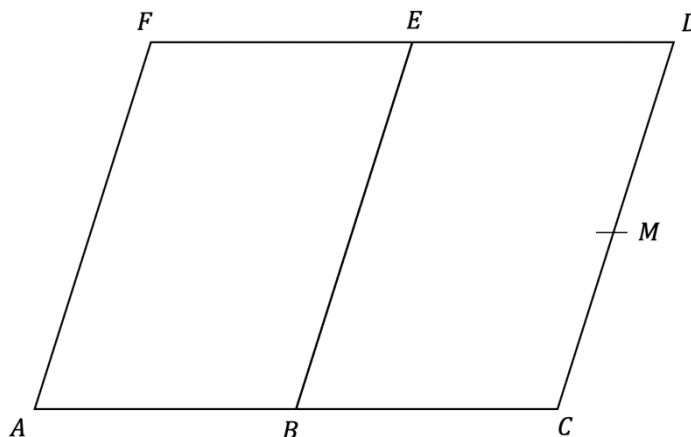
If angle BAC is 101.1° , find the area of triangle ABC ?

- (b) The coefficients of the $(r - 5)^{th}$ and $(2r - 1)^{th}$ terms in the expansion of $(1 + x)^{34}$ are equal. Find the value of r .

3

Question 9 (5 marks) Start a new page.

- (a) The diagram below shows two identical parallelograms, $AFEB$ and $BEDC$. M is the midpoint of CD .



Let the vector $\overrightarrow{AF} = \mathbf{q}$ and $\overrightarrow{AB} = \mathbf{p}$.

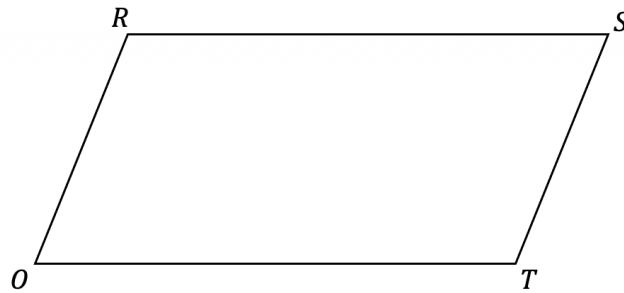
- (i) Use vector methods to show that BM is parallel to AD .
 (ii) Write $\overrightarrow{BM}$ in terms of $\mathbf{p}$ and $\mathbf{q}$.

2

1

Question 9 continues on the next page

- (b) In parallelogram $ORST$, it is known that $\overrightarrow{OR} = \underline{a} + 2\underline{b}$ and that $\overrightarrow{OT} = \underline{a} - 2\underline{b}$, where $\underline{a}$ is the position vector of the point A and $\underline{b}$ is the position vector of the point B . 2



Copy the diagram onto your answer sheet and show the positions of the points A and B .

Question 10 (5 marks) **Start a new page.**

Let X be a continuous random variable and $\lambda > 0$ be a positive real number. X is said to have a Sine Square distribution, if the probability density function of X is expressed as:

$$f_X(x) = \begin{cases} \frac{2}{\lambda\pi} \sin^2 \frac{x}{2\lambda} & \text{for } 0 < x < \lambda\pi, \text{ and } \lambda > 0; \\ 0 & \text{otherwise} \end{cases}$$

Note that $f_X(x) \geq 0$ for all values of X in the domain $(0, \lambda\pi)$.

- (a) Find the cumulative distribution function $F_X(x)$ of the Sine Square distribution. 4
- (b) Hence, or otherwise, find $P(X \leq \pi)$ when $\lambda = 2$. Give your answer to 3 decimal places. 1

Please turn over

Question 11 (5 marks) Start a new page.

(a) Find λ when the magnitude of the vector projection of $\underline{a} = \lambda \underline{i} + \underline{j}$ onto $\underline{b} = 8\underline{i} + 6\underline{j}$ equals 4. 2

(b) By considering the expansion of $(1 + x)^n$, show that 3

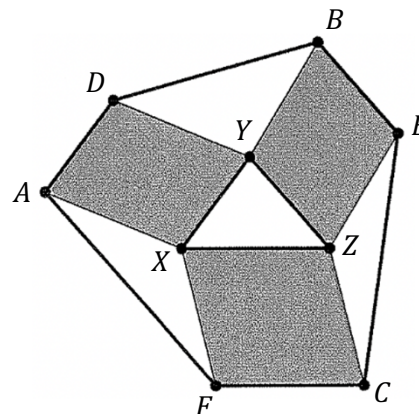
$$2 \binom{n}{2} + 6 \binom{n}{3} + 12 \binom{n}{4} + \dots + n(n-1) \binom{n}{n} = n(n-1)2^{n-2}$$

Question 12 (5 marks) Start a new page.

(a) Suppose that A, B, C, D, E, F are any points in the plane. Use vector arithmetic to verify that 2

$$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{AF}$$

(b) Suppose now that A, B, C, D, E, F are arranged to form the vertices of a hexagon surrounding a triangle XYZ , as shown in the diagram below. It is given that the shaded regions are parallelograms. 3



Show that the line segments DB , EC and FA can be moved parallel to themselves to assemble a triangle.

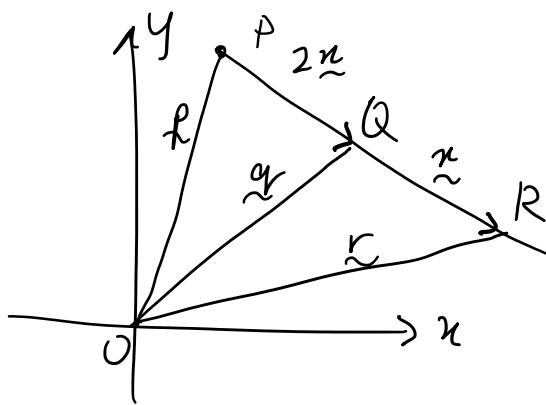
END OF EXAMINATION

Multiple choice

Sunday, 6 June 2021

9:58 pm

1.



$$\begin{aligned} \vec{r} &= \vec{OR} = \vec{OP} + \vec{PQ} + \vec{QR} \\ &= \vec{p} + \vec{PO} + \vec{OQ} + \vec{QR} \\ &= \vec{p} + -\vec{p} + \vec{q} + \frac{1}{2}(-\vec{p} + \vec{q}) \\ &= -\frac{1}{2}\vec{p} + \frac{3}{2}\vec{q} \end{aligned}$$

(A)

2. $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$

$$= 4 \times 4 \times \cos 60^\circ$$

$$= 16 \times \frac{1}{2}$$

$$= 8$$

(D)

3.

~~A~~
(B)
~~C~~
~~D~~

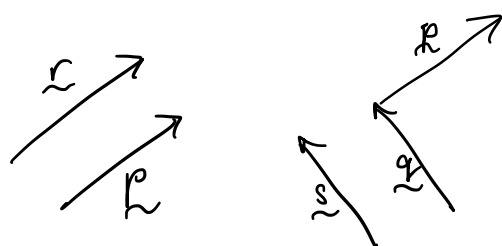
4.

$$\begin{aligned} (1+x)^n (1+\frac{1}{x})^n &= \frac{(1+x)^n (1+x)^n}{x^n} \\ &= \frac{(1+x)^{2n}}{x^n} \end{aligned}$$

$\therefore$ coeff of $\frac{1}{x}$ in $(1+x)^n (1+\frac{1}{x})^n =$ coeff of x^{n-1} in $(1+x)^{2n}$

$$= \binom{2n}{n-1} = \frac{(2n)!}{(n-1)!(n+1)!} \quad \text{(B)}$$

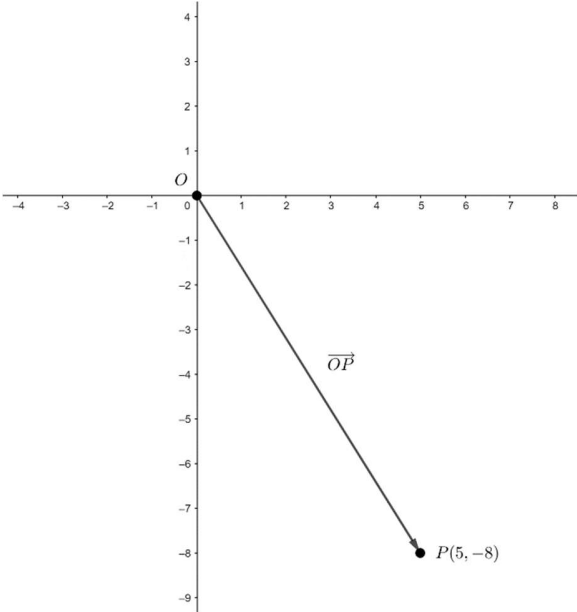
5.



$$\frac{\vec{p} \cdot \vec{q}}{|\vec{q}|^2} \vec{q}$$

(D)

Year 12 Extension 1: Q6

Suggested Solutions	Marks Awarded	Marker's Comments
(a) Subtracting $\overrightarrow{OB}$ from $\overrightarrow{OA}$: (i) $\overrightarrow{AB} = (-5 - 5)\underline{i} + (-2 - (-4))\underline{j} = -10\underline{i} + 2\underline{j}$ (ii) $ \overrightarrow{AB} = \sqrt{(-10)^2 + 2^2}$ $= \sqrt{104}$ $= 2\sqrt{26}$	1 1	Handled well. Handled well.
(b) (i)  (ii) $\overrightarrow{OP} = \begin{bmatrix} 5 \\ -8 \end{bmatrix}$ (iii) Multiply by the reciprocal of the magnitude of $\overrightarrow{OP}$ to obtain a vector in the direction of $\overrightarrow{OP}$ with magnitude 1: $ \overrightarrow{OP} = \sqrt{5^2 + (-8)^2} = \sqrt{25 + 64} = \sqrt{89}$ Hence the required unit vector is $\frac{1}{\sqrt{89}} \begin{bmatrix} 5 \\ -8 \end{bmatrix}$	1 1 1	Some students failed to give reference/place the vector in the Cartesian plane. Handled well, except for notation. Vectors are not points, so rounded brackets shouldn't be used. For example, $(5, -8)$ is notation for a point. Why is this important? Because in the context of vectors, points AND vectors are often mixing together as part of a problem. The two must be separated in notation to avoid confusion.

Question 7 (5 marks)

- a) Find the angle between the vectors $3\mathbf{i} + \mathbf{j}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. (2 marks)
- b) Show that the vector $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$ are perpendicular. (1 mark)
- c) Find a vector of magnitude 15 in the direction of the vector $5\mathbf{i} + 12\mathbf{j}$. (2 marks)

Part (a) Solution

Find the angle between the vectors $3\mathbf{i}+\mathbf{j}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$.

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \left| \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right| \left| \begin{pmatrix} 2 \\ -1 \end{pmatrix} \right| \cos \theta$$

(1 mark)

$$5 = \sqrt{10} \times \sqrt{5} \cos \theta$$

$$\cos \theta = \frac{\sqrt{5}}{\sqrt{10}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

(1 mark)

Part (b) Solution

Show that the vector $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$ are perpendicular.

$$\begin{aligned} \begin{pmatrix} 1 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -2 \end{pmatrix} &= 8 - 8 \\ &= 0 \end{aligned}$$

Since the dot product is 0, the vectors are perpendicular.

(1 mark)

Part (c) Solution

Find a vector of magnitude 15 in the direction of the vector $5\mathbf{j} + 12\mathbf{j}$.

$$|5\mathbf{j} + 12\mathbf{j}| = 13 \quad (1 \text{ mark})$$

$$\begin{array}{l} \text{The vector of magnitude 15} \\ \text{and in the direction of vector} \\ 5\hat{\mathbf{i}} + 12\hat{\mathbf{j}} \end{array} = \frac{15}{13}(5\mathbf{j} + 12\mathbf{j}) \quad (1 \text{ mark})$$

MATHEMATICS Extension 1 : Question... 8....

Suggested Solutions

Marks

Marker's Comments

a)

$$|\vec{AB}| = \sqrt{(-2)^2 + (7)^2} = \sqrt{53}$$

$$|\vec{AC}| = \sqrt{12^2 + (-1)^2} = \sqrt{145}$$

$$\therefore \text{Area} = \frac{\sqrt{53} \times \sqrt{145} \times \sin 101.1^\circ}{2}$$

$$= 43.01 \text{ u}^2$$

①

①

$$b) (1+x)^{34} = {}^{34}C_0 + {}^{34}C_1 x + \dots + {}^{34}C_r x^r + \dots + {}^{34}C_{34} x^{34}$$

$\uparrow$ 1st term $\uparrow$ 2nd term

$$\therefore \text{Coefficient of } (r-5)^{\text{th}} \text{ term} = {}^{34}C_{r-6}$$

$$\text{coefficient of } (2r-1)^{\text{th}} \text{ term} = {}^{34}C_{2r-2}$$

$$\therefore r-6 = 2r-2 \text{ or } (r-6) + (2r-2) = 34$$

$$r = -4$$

$$3r - 8 = 34$$

$$\text{but } r \geq 0.$$

$$3r = 42$$

$$r = 14$$

①

①

①

$$\therefore r = 14 \text{ only.}$$

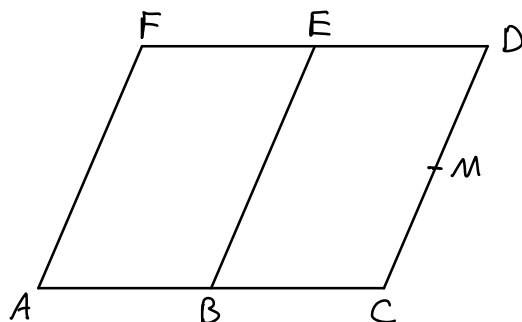
Note: If your answer is $\frac{40}{3}$, you will receive 1/3 MAX!

- Getting $r=14$ is only worth 1 mark on its own without appropriate working

- You will also not receive full marks if your final step was trial and error but you did not sub the value to check that it is correct.

Question 9:

ai,



$$\vec{AF} = z$$

$$\vec{AB} = r$$

RTP: $AD \parallel BM$

$$\vec{AD} = \vec{AC} + \vec{CD}$$

$$= 2\vec{AB} + \vec{CD}$$

$$\vec{AD} = 2r + z$$

①

$$\vec{BM} = \vec{BC} + \vec{CM}$$

$$\vec{BM} = r + \frac{1}{2}z$$

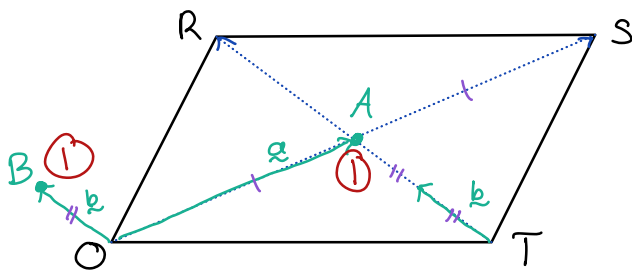
$$= \frac{1}{2}(2r + z)$$

$$\textcircled{1} \left\{ \begin{array}{l} \vec{BM} = \frac{1}{2}\vec{AD} \\ \therefore \vec{BM} \text{ is a scalar multiple of } \vec{AD}, \text{ then } BM \parallel AD. \end{array} \right.$$

ii, From above, $\vec{BM} = r + \frac{1}{2}z$

①

b,



$$\vec{OR} = r + 2b \quad \textcircled{1}$$

$$\vec{OT} = r - 2b \quad \textcircled{2}$$

$$\textcircled{1} + \textcircled{2}: 2a = \vec{OR} + \vec{OT}$$

$$a = \frac{1}{2}(\vec{OR} + \vec{OT})$$

$$\textcircled{1} - \textcircled{2}: 4b = \vec{OR} - \vec{OT}$$

$$b = \frac{1}{4}(\vec{OR} - \vec{OT})$$

MATHEMATICS Extension 1 : Question...10...

Suggested Solutions	Marks	Marker's Comments
a) $\int_0^x \frac{2}{\lambda\pi} \sin^2 \frac{x}{2\lambda} dx$ $= \frac{2}{\lambda\pi} \int_0^x \frac{1}{2} (1 - \cos \frac{x}{\lambda}) dx$ $= \frac{1}{\lambda\pi} \int_0^x (1 - \cos \frac{x}{\lambda}) dx$ $= \frac{1}{\lambda\pi} \left[x - \lambda \sin \frac{x}{\lambda} \right]_0^x$ $= \frac{1}{\lambda\pi} \left[x - \lambda \sin \frac{x}{\lambda} \right] - 0$ $\therefore$ $F_x(x) = \begin{cases} 0, & x \leq 0 \\ \frac{1}{\lambda\pi} \left(x - \lambda \sin \frac{x}{\lambda} \right), & 0 < x < \lambda\pi \\ 1, & x \geq \lambda\pi \end{cases}$	1 1 1 1	setting up the integral with boundaries. If not using boundaries, first mark will go towards working out the + c. must show all three parts. Not many students did this.
b) when $\lambda=2$, $x=\pi$ $P(x \leq \pi) = \frac{1}{2\pi} \left(\pi - 2 \sin \frac{\pi}{2} \right)$ $= \frac{\pi}{2\pi} - \frac{2}{2\pi}$ $= \frac{1}{2} - \frac{1}{\pi}$ $= 0.182 \text{ (3dp)}$	1	$\frac{1}{2} - \frac{1}{\pi}$ was also accepted. The sin value must be evaluated. A lot of students used degrees rather than radians!

Question 11

Sunday, 8 August 2021 10:15 pm

(a) Find λ when the magnitude of the vector projection of $\underline{a} = \lambda\hat{i} + \hat{j}$ onto $\underline{b} = 8\hat{i} + 6\hat{j}$ equals 4.

2

Solution:

$$\begin{aligned} |\text{proj}_{\underline{b}} \underline{a}| &= \left| \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|} \right| \\ &= \left| \frac{\begin{pmatrix} \lambda \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 6 \end{pmatrix}}{\sqrt{8^2 + 6^2}} \right| \\ &= \left| \frac{8\lambda + 6}{10} \right| = 4 \end{aligned}$$

$$8\lambda + 6 = 40 \quad \text{or} \quad 8\lambda + 6 = -40$$

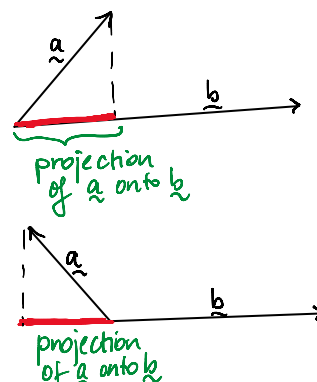
$$8\lambda = 34 \quad \quad \quad 8\lambda = -46$$

$$\lambda = 4.25 \quad \text{or} \quad -5.75$$

①

①

Many student did not consider the magnitude of the vector projection. They omitted the absolute value and so only found one value of λ .



1 mark was awarded for only one value of λ .

2 marks were awarded for two correct values of λ .

(b) By considering the expansion of $(1+x)^n$, show that

3

$$2 \binom{n}{2} + 6 \binom{n}{3} + 12 \binom{n}{4} + \dots + n(n-1) \binom{n}{n} = n(n-1)2^{n-2}$$

Solution:

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \binom{n}{4}x^4 + \dots + \binom{n}{n}x^n$$

Differentiating both sides:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + 4\binom{n}{4}x^3 + \dots + n\binom{n}{n}x^{n-1}$$

Differentiating both sides again:

$$n(n-1)(1+x)^{n-2} = 2\binom{n}{2} + 6\binom{n}{3}x + 12\binom{n}{4}x^2 + \dots + n(n-1)\binom{n}{n}x^{n-2}$$

Substituting $x=1$ in both sides

$$n(n-1)2^{n-2} = 2\binom{n}{2} + 6\binom{n}{3} + 12\binom{n}{4} + \dots + n(n-1)\binom{n}{n}$$

1st mark was awarded for differentiating both sides once with explanation

2nd mark was awarded for differentiating both sides again with explanation

3rd mark was awarded for substitution of $x=1$ into the expression with explanation

Notes:

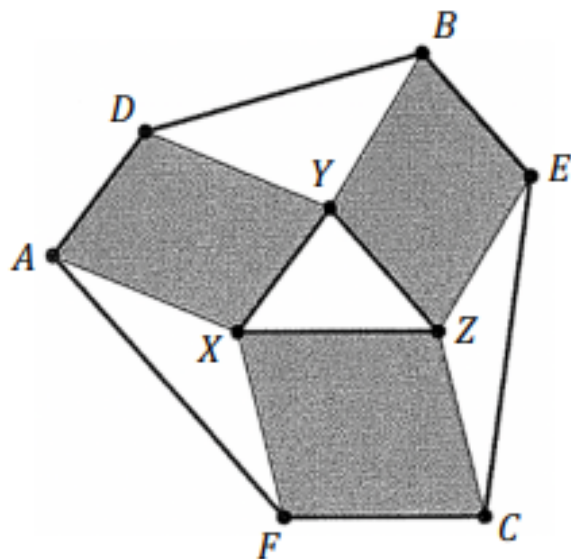
- Many students did not write out enough terms in their initial statement so were unable to show the final statement given in the question.
- Students wrote the extra terms in subsequent lines of working without showing where they initially came from.
- Students who did not show enough terms to arrive at the given expression were awarded maximum 2 marks
- Many students wrote lines of working without explaining how they arrived at each line.
- Students who did not explain they were differentiating each line were awarded only the mark for substitution.
- Many students who used unacceptable notation and explanations were NOT awarded marks. These included: LHS', RHS', $\frac{d}{dx}$, $((1+x)^n)'$, "diff"

Question 12 (5 marks) **Start a new page.**

- (a) Suppose that A, B, C, D, E, F are any points in the plane. Use vector arithmetic to verify that **2**

$$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{AF}$$

- (b) Suppose now that A, B, C, D, E, F are arranged to form the vertices of a hexagon surrounding a triangle XYZ , as shown in the diagram below. It is given that the shaded regions are parallelograms. **3**



Show that the line segments DB , EC and FA can be moved parallel to themselves to assemble a triangle.

(a) Suppose that A, B, C, D, E, F are any points in the plane. Use vector arithmetic to verify that

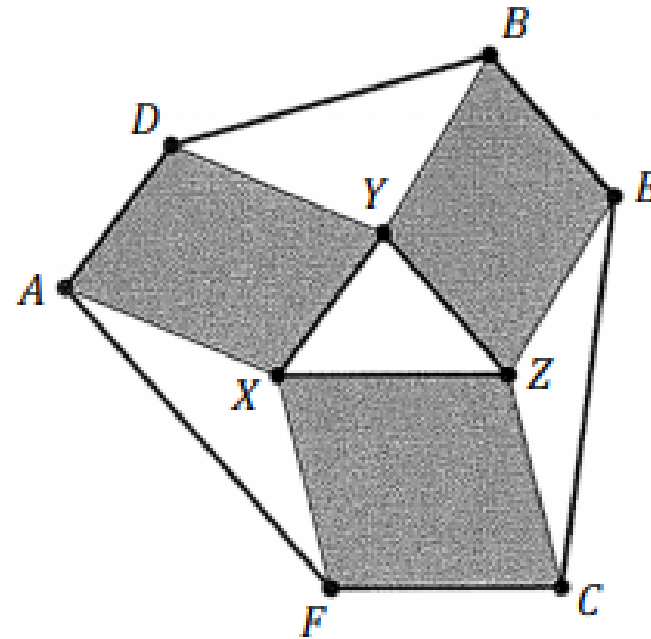
2

$$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} = \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{AF}$$

$$\begin{aligned} & \overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF} \\ = & \overrightarrow{AF} + \overrightarrow{FD} + \overrightarrow{BD} + \overrightarrow{DE} + \overrightarrow{CE} + \overrightarrow{EF} & 1 \\ = & \overrightarrow{AF} + \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{EF} + \overrightarrow{FD} + \overrightarrow{DE} \\ = & \overrightarrow{AF} + \overrightarrow{BD} + \overrightarrow{CE} + \overrightarrow{EE} & 1 \\ = & \overrightarrow{AF} + \overrightarrow{BD} + \overrightarrow{CE} \end{aligned}$$

- (b) Suppose now that A, B, C, D, E, F are arranged to form the vertices of a hexagon surrounding a triangle XYZ , as shown in the diagram below. It is given that the shaded regions are parallelograms.

3



Show that the line segments DB , EC and FA can be moved parallel to themselves to assemble a triangle.

$$\begin{aligned} & \overrightarrow{BD} + \overrightarrow{AF} + \overrightarrow{CE} \\ = & \overrightarrow{BY} + \overrightarrow{YD} + \overrightarrow{AX} + \overrightarrow{XF} + \overrightarrow{CZ} + \overrightarrow{ZE} \end{aligned} \quad 1$$

$\overrightarrow{BY} = \overrightarrow{EZ}$ Since BYZE is a parallelogram, and opposite sides are parallel and equal.

$\overrightarrow{YD} = \overrightarrow{XA}$ Since ADYX is a parallelogram, and opposite sides are parallel and equal 1

$\overrightarrow{XF} = \overrightarrow{ZC}$ Since XFCZ is a parallelogram, and opposite sides are parallel and equal

Now using The above informations, we obtain:

$$\begin{aligned} & \overrightarrow{BY} + \overrightarrow{YD} + \overrightarrow{AX} + \overrightarrow{XF} + \overrightarrow{CZ} + \overrightarrow{ZE} \\ = & \overrightarrow{BY} + \overrightarrow{YD} + \overrightarrow{XF} - \overrightarrow{YD} - \overrightarrow{XF} - \overrightarrow{BY} \\ = & \overrightarrow{BY} - \overrightarrow{BY} + \overrightarrow{YD} - \overrightarrow{YD} + \overrightarrow{XF} - \overrightarrow{XF} \\ = & \vec{0} \end{aligned} \quad 1$$

Therefore the three vectors $\overrightarrow{DB}$, $\overrightarrow{EC}$ and $\overrightarrow{FA}$ are the sides of a triangle.